

**AP PHYSICS C: ELECTRICITY
AND MAGNETISM**

UNIT 2

**Conductors,
Capacitors,
Dielectrics**



14–17%

AP EXAM WEIGHTING

UNIT AT A GLANCE

Enduring Understanding	Topic	Suggested Skills
ACT-2, ACT-3	2.1 Conductors, Capacitors, Dielectrics: Electrostatics with Conductors	1.A Describe the physical meaning (includes identifying features) of a representation. 1.E Describe the effects of modifying conditions or features of a representation of a physical situation. 5.A Select an appropriate law, definition, mathematical relationship, or model to describe a physical situation. 7.C Support a claim with evidence from physical representations. 7.D Provide reasoning to justify a claim using physical principles or laws.
	2.2 Conductors, Capacitors, Dielectrics: Capacitors	1.A Describe the physical meaning (includes identifying features) of a representation. 2.A Identify a testable scientific question or problem. 2.B Make a claim or predict the results of an experiment. 2.E Identify or describe potential sources of experimental error. 5.B Determine the relationship between variables within an equation when an existing variable changes. 7.A Make a scientific claim. 7.D Provide reasoning to justify a claim using physical principles or laws.
CNV-4		

continued on next page

UNIT AT A GLANCE *(cont'd)*

Enduring Understanding	Topic		Suggested Skills	
FIE-2	2.3 Conductors, Capacitors, Dielectrics: Dielectrics		2.B	Make a claim or predict the results of an experiment.
			3.C	Sketch a graph that shows a functional relationship between two quantities.
			3.D	Create appropriate diagrams to represent physical situations.
			5.B	Determine the relationship between variables within an equation when an existing variable changes.
			6.C	Calculate an unknown quantity with units from known quantities, by selecting and following a logical computational pathway.
			7.A	Make a scientific claim.
			7.B	Support a claim with evidence from experimental data.

TestDaily



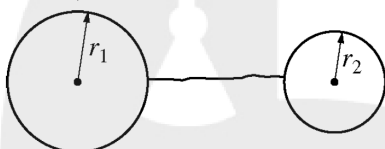
SECTION1- MC 真题练习

1. (2012-5)

All the following statements about an isolated, solid charged conductor are correct EXCEPT:

- (A) All parts of the conductor are at the same potential.
- (B) All excess charge resides on the outer surface.
- (C) The net charge enclosed by any surface lying entirely within the conductor must equal zero.
- (D) The electric field $\mathbf{E}$ just outside the conductor is directed parallel to the surface.
- (E) The electric field intensity inside the conductor is zero.

2. (2012-14)

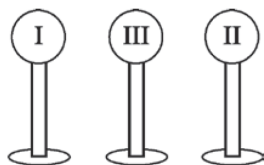


Note: Figure not drawn to scale.

A metal sphere with radius r_1 has a total electric charge of magnitude q . An uncharged metal sphere with radius r_2 (with $r_1 > r_2$) is then connected by a wire to the first sphere, as illustrated above. The separation of the spheres is much greater than the radius of either sphere. When equilibrium is reached, the spheres will have

- (A) charges on their surfaces of equal magnitude and the same sign
- (B) charges on their surfaces of equal magnitude and opposite sign
- (C) equal electric fields at their surfaces
- (D) equal capacitances
- (E) equal electric potentials

3. (2013-14)



Three identical conducting spheres are mounted on insulating handles, as shown above. Spheres I and II have equal charges of $+Q$ and are separated by a fixed distance. They repel each other with an electrostatic force of magnitude F . Sphere III, initially uncharged, is first touched to sphere I, then to sphere II, and then removed. If the charge distribution on each sphere is assumed to always be spherical, the new magnitude of the electrostatic force between spheres I and II is

- (A) zero
- (B) $\frac{F}{16}$
- (C) $\frac{F}{4}$
- (D) $\frac{F}{2}$
- (E) $\frac{3F}{8}$

4. (2013-15)

Which of the following describes the electric field and electric potential inside a charged solid spherical conductor that is in electrostatic equilibrium?

<u>Field</u>	<u>Potential</u>
(A) Zero	Zero
(B) Nonzero and uniform	Zero
(C) Nonzero and uniform	Nonzero and uniform
(D) Zero	Nonzero and uniform
(E) Zero	Nonzero and nonuniform

5. (2014-33)

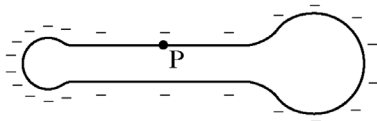
A solid metal sphere is in equilibrium and has a net charge Q placed on it. If the sphere is heated so that it expands uniformly without affecting the amount of charge and so that it is still in equilibrium, which of the following will be unaffected by the expansion?

- I. The surface charge density
- II. The electric potential inside the sphere
- III. The electric field inside the sphere

- (A) I only
- (B) II only
- (C) III only
- (D) I and II
- (E) II and III



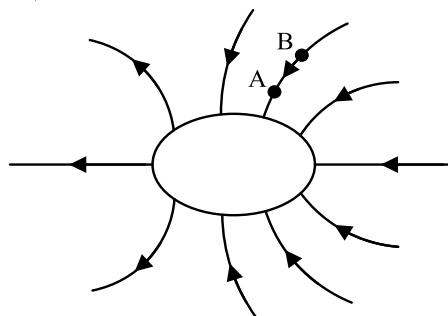
6. (2015-15)



The conductor shown above is given a negative charge that spreads over its surface. Which of the following best represents the direction of the electric field at the point P on the cylindrical portion of the conductor?

- (A) (B) (C) (D) (E)

7. (2021-DE-D61102-Section 1-15)

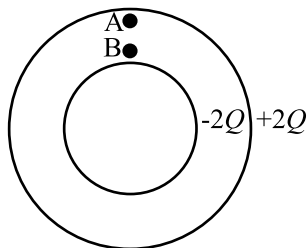


The figure shows the electric field lines around an object of ellipsoidal shape. Which of the following must be true?

- A. The object is a solid conductor.
- B. The object is hollow conductor.
- C. The object has zero net charge.
- D. The electric field is larger at point A than at point B.
- E. The electric field has the same magnitude at point A than at point B.

TestDaily

8. (2021-DE-D61102-Section 1-14)



Two concentric conducting shells form a spherical capacitor with charge of magnitude $2Q$ on each plate, as shown in the figure. Which of the following sketches best depicts the direction and relative magnitude of the electric field vectors at points A and B?

A.

Electric Field Vector at Point A	Electric Field Vector at Point B

B.

Electric Field Vector at Point A	Electric Field Vector at Point B

C.

Electric Field Vector at Point A	Electric Field Vector at Point B

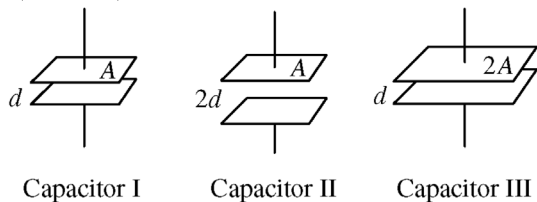
D.

Electric Field Vector at Point A	Electric Field Vector at Point B

E.

Electric Field Vector at Point A	Electric Field Vector at Point B

9. (2012-10)

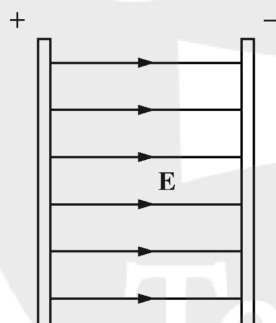


The plate areas and separations for three capacitors are shown in the diagram above. The space between the plates in each capacitor is filled with air.

Suppose all three capacitors have charge $+Q$ on the top plate and charge $-Q$ on the bottom plate. Which of the following is true of the potential difference across the plates of the three capacitors?

- (A) It is greatest for I.
- (B) It is greatest for II.
- (C) It is greatest for III.
- (D) It is the same for II and III and least for I.
- (E) It is the same for all three capacitors.

10. (2012-20)



A uniform electric field E exists between the two large, oppositely charged plates shown above.

If the distance between the plates is increased without changing the charges on the plates, which of the following statements can be justified?

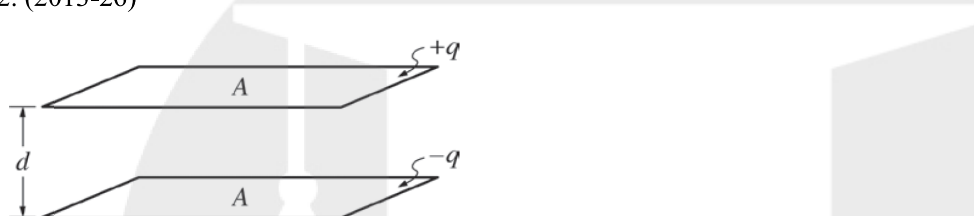
- (A) The electric field strength decreases.
- (B) The electric field strength increases.
- (C) The potential difference between the plates decreases.
- (D) The potential difference between the plates increases.
- (E) There will be no change in either the electric field strength or the potential difference.

11. (2013-16)

A parallel-plate capacitor is connected to a battery until fully charged. If the battery remains connected while the separation between the plates is increased, which of the following remains constant?

- (A) Potential difference between the plates
- (B) Capacitance
- (C) Magnitude of the charge on each plate
- (D) Stored electrostatic energy
- (E) Electric field intensity in the region between the plates

12. (2013-26)



A parallel-plate capacitor of capacitance C consists of two plates of area A separated by distance d as shown above. The upper and lower plates are given a net charge of $+q$ and $-q$, respectively. What is the electric field between the plates?

- (A) Zero
- (B) $\frac{C}{2qd}$
- (C) $\frac{C}{qd}$
- (D) $\frac{2q}{Cd}$
- (E) $\frac{q}{Cd}$

13. (2013-27)

Five air-filled parallel-plate capacitors have the plate areas and plate separations listed below, where A and d are constants. The capacitors are each connected to the same potential difference. Which capacitor stores the greatest amount of energy?

	<u>Area</u>	<u>Separation</u>
(A)	$2A$	$d/2$
(B)	$2A$	$2d$
(C)	A	d
(D)	$A/2$	$d/2$
(E)	$A/2$	$2d$

14. (2014-14)

The maximum charge a capacitor can store on one plate is limited by which of the following?

- (A) How much charge can physically fit on the conducting plates
- (B) The maximum time rate of change of the charge on the other plate
- (C) The nonzero energy needed to remove an electron from the conducting plates
- (D) The finite value of the capacitance
- (E) The electrical discharge when the electric field between the plates becomes too great

15. (2014-16)

Circuit P consists of three identical capacitors connected in parallel with a battery. Circuit S consists of the same three capacitors connected in series with the same battery. When the capacitors are fully charged, what is the ratio of the total energy stored in circuit P to the total energy stored in circuit S ?

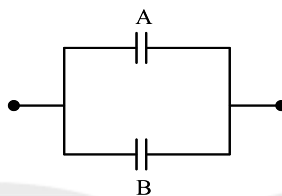
- (A) 9
- (B) 3
- (C) 1
- (D) $1/3$
- (E) $1/9$

16. (2014-32)

If the charge on a parallel-plate capacitor is decreased from 6 pC to 2 pC and the plate separation is increased from 1 mm to 3 mm, the energy stored in the capacitor will change from U_0 to

- (A) $U_0/4$
- (B) $U_0/3$
- (C) $3U_0$
- (D) $8U_0$
- (E) $27U_0$

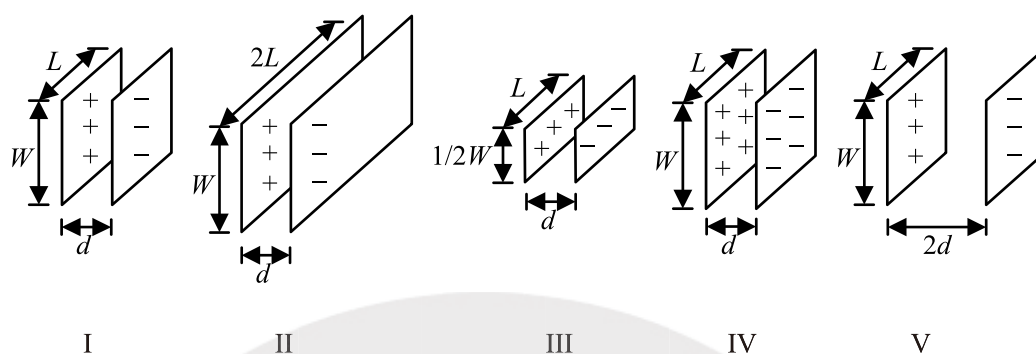
17. (2021-DE-D61102-Section 1-1)



Two capacitors, A and B, having capacitances of C_A and C_B , respectively, are connected as shown. A potential difference is applied across the terminals, and the total energy stored in the pair of capacitors is $3 \times 10^{-6} \text{ J}$. A dielectric is inserted into capacitor B with a dielectric constant value of 2. With the same potential difference applied, the total energy stored in the pair of capacitors is $5 \times 10^{-6} \text{ J}$. What is the ratio of the capacitances, $\frac{C_B}{C_A}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. $\frac{5}{3}$
- E. 2

18. (2021-DE-D61102-Section 1-16)

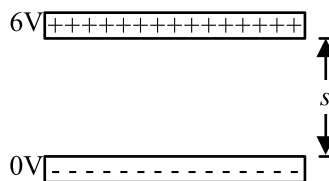


The figure shows five isolated, charged parallel-plate capacitors. Capacitor I has a charge of $+3Q$ on its positive plate, length L , width W , and separation between the plates d . Capacitor II is similar to capacitor I but the length of the plates is twice as long. Capacitor III's plates are half as wide as those of capacitor I. Capacitor IV has twice the charge that capacitor I has on its positive plate. The plate separation of capacitor V is twice that of capacitor I. Which of the following correctly ranks the capacitors by electric field strength, from strongest to weakest?

- A. $I > IV = V > III = II$
- B. $II > I = IV = V > III$
- C. $III = IV > I = V > II$
- D. $IV > I = V > II > III$
- E. $V > IV > I > II > III$

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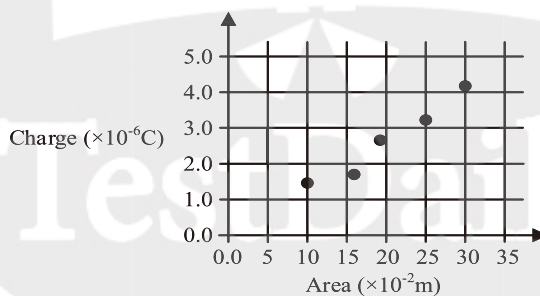
19. (2021-DE-D61102-Section 2-4)



The figure shows a parallel plate capacitor with a charge of magnitude $1.0 \mu\text{C}$ on each plate, a cross sectional area of 0.20m^2 , and a small distance of separation of s between them. The potential difference between the plates is 6.0 V . The magnitude and the direction of the electric field of the capacitor is given by

- A. $1.1 \times 10^5 \text{ N/C}$ downward
- B. $1.1 \times 10^5 \text{ N/C}$ upward
- C. $5.7 \times 10^5 \text{ N/C}$ downward
- D. $5.7 \times 10^5 \text{ N/C}$ upward
- E. $3.4 \times 10^6 \text{ N/C}$ upward

20. (2021-DE-D61102-Section 2-9)



Five air-filled parallel plate capacitors with the same separation distance but different plate areas are each connected to a 1.5V battery. The charge on one of the positive plates as a function of plate area is represented in the graph shown. The distance between the plates of the capacitor is most nearly

- A. $1.0 \times 10^{-6} \text{ m}$
- B. $1.1 \times 10^{-3} \text{ m}$
- C. $1.8 \times 10^{-14} \text{ m}$
- D. $4.4 \times 10^{-7} \text{ m}$
- E. $1.1 \times 10^5 \text{ m}$

21. (2021-DE-D62503-Section 1-1)

An air-filled parallel plate capacitor with circular plates of radius R has a total charge magnitude of Q distributed uniformly on each plate. The electric field between the plates is E . A second air-filled parallel plate capacitor has a radius of $2R$ and a charge of $4Q$. The electric field between the plates of the second capacitor is

A. $4E$

B. $2E$

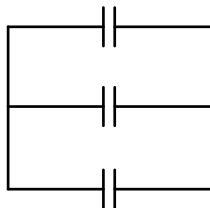
C. E

D. $\frac{E}{2}$

E. $\frac{E}{4}$



22. (2021-DE-D62503-Section 2-7)



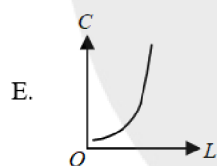
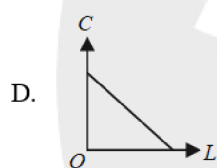
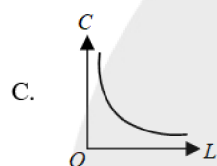
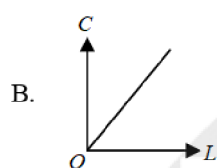
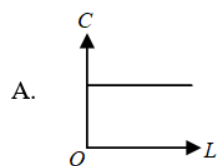
One of the three identical parallel plate capacitors is given a charge, q , so that it has energy U . The other two capacitors are initially uncharged. The capacitors are then connected as shown in the figure. After steady-state is reached, the total energy of the new system of capacitors is

A. $3U$ B. U C. $\frac{U}{3}$ D. $\frac{U}{6}$ E. $\frac{U}{9}$

23. (2021-DE-D62503-Section 2-22)

A set of capacitors are constructed with square plates of plate separation d and varying side lengths L .

Which graph represents how the capacitance C varies as a function of L ?



24. (2013-25)

A parallel-plate capacitor is charged and the battery is removed. If a dielectric with dielectric constant $k > 1$ is then inserted in the capacitor, which of the following will decrease?

- (A) The voltage across the capacitor
- (B) The charge on one of the capacitor plates
- (C) The distance between the capacitor plates
- (D) The capacitance of the capacitor
- (E) The resistance of the capacitor

25. (2015-14)

If a dielectric is inserted between the plates of a capacitor while the capacitor maintains its connection to a constant voltage source, which of the following is true?

- (A) The capacitance of the capacitor is unchanged.
- (B) The charge on the capacitor plates increases.
- (C) The potential difference across the capacitor increases.
- (D) The electric field between the capacitor plates increases.
- (E) The electric field between the capacitor plates decreases.

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26. (2021-DE-D61102-Section 1-19)

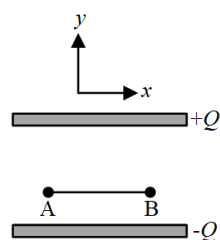


Figure 1

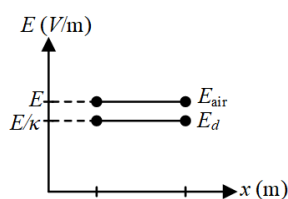
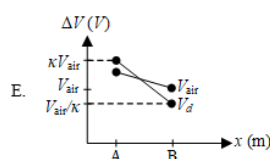
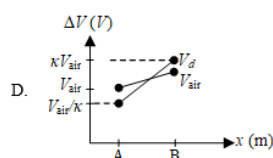
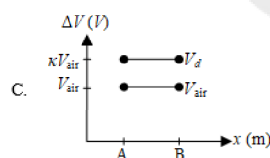
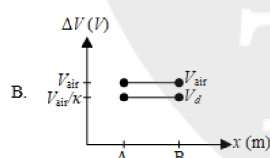
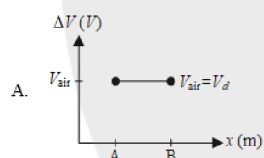


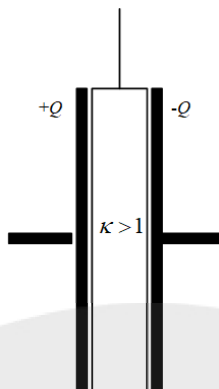
Figure 2

A parallel plate capacitor is charged and insulated from its surroundings such that the upper and lower plates have charges $+Q$, and $-Q$, respectively, as shown in Figure 1. Later, a dielectric material of dielectric constant κ is inserted in the space between the plates. Figure 2 shows the graph of the magnitude of the electric field E along the path AB before and after the dielectric material is inserted, labeled E_{air} , and E_d , respectively.

Which of the following graphs correctly shows the potential difference ΔV_{air} and ΔV_d between the bottom plate and the points along the path AB as a function of position, before and after the dielectric is inserted, respectively?



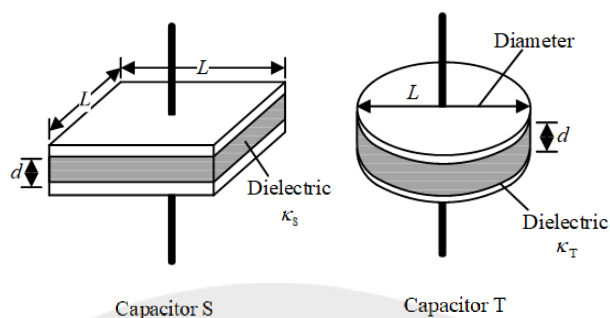
27. (2021-DE-D61102-Section 1-29)



A parallel plate capacitor has a slab of dielectric material with $\kappa > 1$ in its center supported by a string of negligible mass, as shown. The capacitor plates are held in place with clamps and will not move. The capacitor is charged by a battery until the charges on the two plates are $+Q$ and $-Q$, and then the battery is removed. The string is cut, allowing the dielectric slab to fall downward. Assume friction and air resistance are negligible and the experiment is done at Earth's surface. The dielectric slab accelerates with an acceleration a , where a is

- A. equal to g , since the capacitor does not exert a net force on the dielectric
- B. smaller than g , since the total charge on the capacitor must increase, and this requires energy
- C. smaller than g , since the potential energy of the capacitor will increase
- D. greater than g , since the potential difference drops, so energy is released from the capacitor to the dielectric
- E. greater than g , since the total charge on the capacitor will decrease, and this releases energy

28. (2021-DE-D61102-Section 2-20)



Two capacitors of the same capacitance C have the identical plate separation d . Capacitor S has square plates of side L and is filled with a material of dielectric constant κ_S . Capacitor T has circular plates of diameter L and is filled with a material of dielectric constant κ_T . Which of the following could represent the relationship between κ_S and κ_T ?

- A. $\kappa_T = \kappa_S$
- B. $\kappa_T = \frac{4}{\pi} \kappa_S$
- C. $\kappa_T = \frac{\pi}{4} \kappa_S$
- D. $\kappa_T = \frac{1}{\pi} \kappa_S$
- E. $\kappa_T = \pi \kappa_S$

29. (2021-DE-D62503-Section 2-4)

An air-filled $5\mu F$ capacitor is connected to a 10 V battery and fully charged. The capacitor is then disconnected and isolated from the battery. The plates of the capacitor are pulled apart so that the distance between the plates is increased to 125% of the original value. Dielectric material ($\kappa = 2$) is then inserted between the plates so that it fills the entire space between the plates. What is the change in capacitance of the capacitor?

A. $\Delta C = +3\mu F$

B. $\Delta C = +1\mu F$

C. $\Delta C = 0$

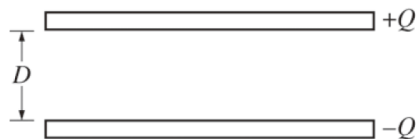
D. $\Delta C = -1\mu F$

E. $\Delta C = -3\mu F$

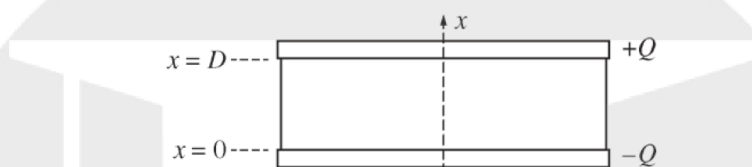


SECTION2- FRQ 真题练习

30. (2015-1b/d/e)



A parallel-plate capacitor is constructed of two parallel metal plates, each with area A and separated by a distance D . The plates of the capacitor are each given a charge of magnitude Q , as shown in the figure above. Ignore edge effects.



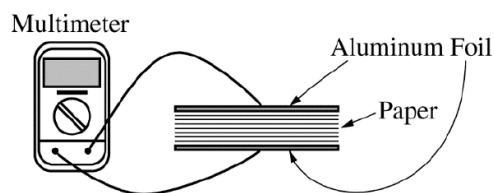
The space between the plates is now filled with a dielectric material that is engineered so that its dielectric constant varies with the distance from the bottom plate to the top plate, defined by the x -axis indicated in the diagram above. As a result, the electric field between the plates is given by $\vec{E} = -\frac{Q}{\epsilon_0 \kappa_0 e^{-x/D} A} \hat{i}$, where κ_0 is a positive constant. Express all algebraic answers to the remaining parts in terms of A , D , Q , κ_0 , x , and physical constants, as appropriate.

- (b) Determine an expression for the dielectric constant κ as a function of x .
- (d) Determine the capacitance of the capacitor.
- (e) The energy stored in the capacitor that has a varying dielectric is U_V . A second capacitor that has a constant dielectric of value κ_0 is also given a charge Q . The energy stored in the second capacitor is U_C . How do the values of U_V and U_C compare?

_____ $U_V < U_C$ _____ $U_V > U_C$ _____ $U_V = U_C$

Justify your answer.

31. (2018-2a~c)



An experiment is designed to measure the dielectric constant of paper that has an area $A = 0.060 \text{ m}^2$. Using aluminum foil, two parallel plates are created with the same area as the paper. Five hundred sheets of paper are placed between the aluminum foil plates to create a parallel plate capacitor, as shown in the figure above. Using a multimeter, the capacitance C of the capacitor is measured. The number of sheets and the total thickness d of the stack of paper are recorded. The experiment is repeated, reducing the number of sheets of paper each time. The data are recorded in the table below.

Sheets of Paper	d (m)	C (F)		
500	0.045	6.5×10^{-11}		
400	0.036	7.4×10^{-11}		
300	0.027	8.9×10^{-11}		
200	0.018	11.9×10^{-11}		
100	0.010	21.0×10^{-11}		

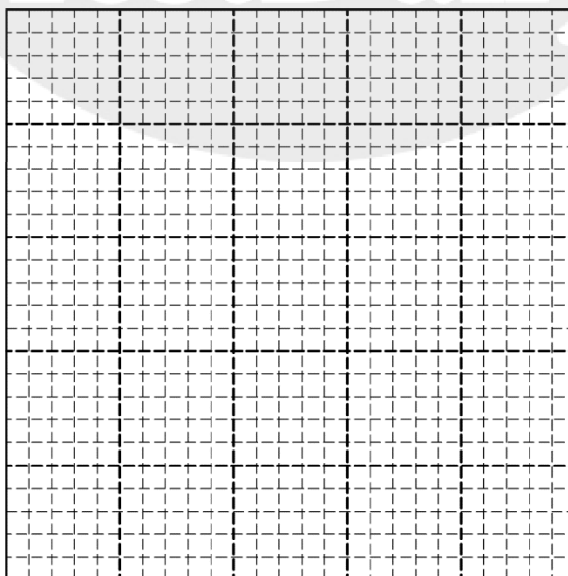
- (a) Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the dielectric constant of the paper.

Vertical axis: _____

Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

- (b) Plot the data points for the quantities indicated in part (a) on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.



- (c) Using the straight line, calculate a dielectric constant for the paper.



**Answer Key for AP Physics C:Electricity and
Magnetism Practice Book, Unit 2**

Question 1: D
Question 2: E
Question 3: E
Question 4: D
Question 5: C
Question 6: C
Question 7: D
Question 8: D
Question 9: B
Question 10: D
Question 11: A
Question 12: E
Question 13: A
Question 14: E
Question 15: A
Question 16: B
Question 17: E
Question 18: C

Question 19: C
Question 20: A
Question 21: C
Question 22: C
Question 23: E
Question 24: A
Question 25: B
Question 26: B
Question 27: C
Question 28: B
Question 29: A

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30.

Distribution
of points

(b) 1 point

Comparing the equation for electric field between parallel plates to the given equation:

$$E = \frac{Q}{\kappa \epsilon_0 A} = \frac{Q}{\epsilon_0 \kappa_0 e^{-x/D} A}$$

For an answer consistent with part (a)-iii

$$\kappa = \kappa_0 e^{-x/D}$$

1 point

(d) 1 point

Using the equation for capacitance:

$$C = \frac{Q}{\Delta V} = \frac{Q}{\left(\frac{QD}{\epsilon_0 \kappa_0 A}\right)(e-1)}$$

For an answer consistent with part (c)-ii

1 point

$$C = \frac{\epsilon_0 \kappa_0 A}{D(e-1)}$$

$$C = \frac{\epsilon_0 \kappa_0 A}{1.72D}$$

(e) 3 points

For selecting $U_V > U_C$

1 point

For correctly comparing the capacitance or the potential difference with the varying dielectric constant to the capacitance or the potential difference with the uniform dielectric constant

1 point

For correctly comparing the two stored energies consistent with the comparison of the capacitances or potential differences

1 point

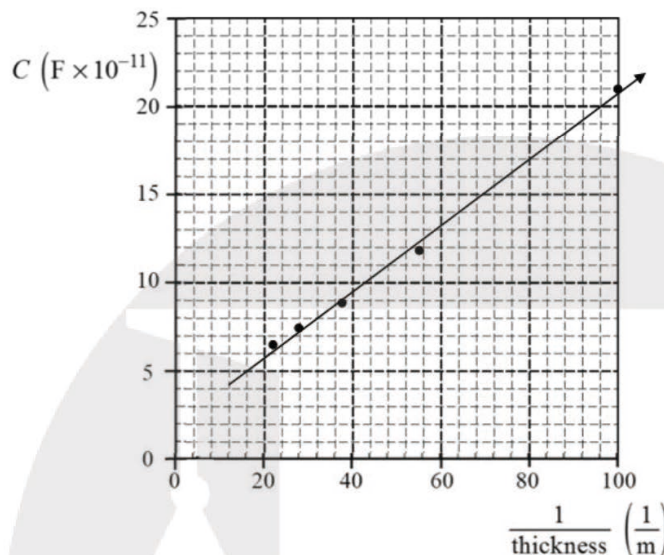
Example: According to the equation from part (d), $C_C > C_V$. Since $U = \frac{Q^2}{2C}$, if the charge stored on the two capacitors is the same, then $U_V > U_C$.

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31.(a) 1 point

For indicating variables that will create a straight line whose slope can be used to determine the dielectric constant of the paper	1 point
Example: Vertical axis: C Horizontal axis: $\frac{1}{d}$	
Note: Student earns full credit if the axes are reversed or if they use another acceptable combination.	

(b) 4 points



For a correct scale that uses more than half the grid	1 point
For correctly labeling the axes including units	1 point
For correctly plotting the data	1 point
For drawing a straight line consistent with the plotted data	1 point

(c) 3 points

For correctly calculating the slope from the best-fit line and not the data points unless the points fall on the best-fit line	1 point
$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{(17 - 8)(F \times 10^{-11})}{(80 - 32)(1/m)} = 0.19 F \cdot m \quad (\text{Linear regression} = 0.187 F \cdot m)$	
For correctly relating the slope to the dielectric constant	1 point
$\text{slope} = \kappa \epsilon_0 A$	
$\kappa = \frac{\text{slope}}{\epsilon_0 A} = \frac{(0.19 F \cdot m)}{(8.85 \times 10^{-12} C^2/N \cdot m^2)(0.060 m^2)}$	
For a correct answer	1 point
$\kappa = 3.58 \quad (\text{Linear regression} = 3.52)$	